INTRODUCTION TO DATA MANAGEMENT
PROJECT REPORT

(Project Semester January-April 2026)

(AGRICULTURAL SALES ANALYSIS DASHBOARD)

Submitted by

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Programme and Section: B.Tech. In CSE, 2M030

Course Code INT387

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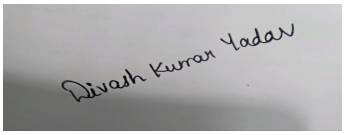
DECLARATION

I, Divash Kumar Yadav, student of B.Tech. in CSE (Program name) under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12/04/2026 Signature



Registration No. 12406109

CERTIFICATE

This is to certify that Divash Kumar Yadav bearing Registration no. 12406109 has completed INT387 project titled, "AGRICULTURAL SALES ANALYSIS DASHBOARD" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Dr. Ashima Bansal

Professor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 20/04/2026

Acknowledgement

I would like to take this opportunity to express my sincere gratitude to all those who contributed to the successful completion of this Agricultural Sales Analysis Dashboard project.

I am thankful for the academic guidance and constructive feedback received throughout the process, which helped improve the quality and depth of the analysis. The support provided through institutional resources, technical tools, and learning opportunities was essential in carrying out this project effectively.

I also acknowledge the valuable dataset used in this project, which enabled accurate and comprehensive insights into agricultural sales trends, product-wise performance, and regional distribution patterns across different parts of India.

I would also like to thank Microsoft Power BI for providing a powerful platform to transform raw data into meaningful visual insights. Working on this project helped me strengthen my skills in data cleaning, data modeling, DAX, and interactive dashboard development.

Lastly, I appreciate the encouragement and support from my family, friends, and mentors throughout the course of this work, which kept me motivated and focused.

Table of Content

• Introduction (page 6)

• Source of Dataset (page 7)

• Data Preprocessing (page 8)

• Objective 1: Overall Sales Performance Analysis (page 10)

• Objective 2: Product-wise Sales Understanding (page 11)

• Objective 3: Region-wise Sales Distribution (page 13)

• Objective 4: Time-based Trend Analysis (page 15)

• Objective 5: Interactive & Detailed Data Exploration (page 17)

• Conclusion (page 19)

• Future Scope (page 20)

• References (page 21)

• Dashboard (Appendix / LinkedIn Post) (page 22)

• GitHub REPOSITORY (URL) (page 24)

Introduction

The Agricultural Sales Analysis Dashboard project is designed to analyze and visualize agricultural sales data across different products, regions, and time periods using Microsoft Power BI. The main objective of this project is to transform raw sales data into meaningful insights through interactive and user-friendly dashboards.

This project focuses on understanding key aspects of agricultural commerce, such as total sales performance, product-wise contribution, region-wise distribution, and time-based trends. By using charts, maps, KPIs, and filters, the dashboard helps identify patterns, compare performance across different products and regions, and support better decision-making.

The dataset used in this project contains agricultural sales information covering multiple products such as Rice, Onion, Brinjal, Cabbage, and Groundnut Oil (Packed), along with sales records across various Indian cities and regions.

Through this project, I enhanced my skills in data cleaning, data transformation, dashboard design, DAX functions, and data storytelling. This project demonstrates the practical application of business intelligence tools in solving real-world analytical problems and presenting insights in an effective manner.

Source of dataset

The dataset utilized in this project contains agricultural sales transaction data covering multiple products and regions across India. It includes key metrics such as product names, regions (cities), sales quantities, prices, and date of sale, enabling a comprehensive analysis of the agricultural market.

Dataset Overview:

- Products Covered: Rice, Onion, Brinjal, Cabbage, Groundnut Oil (Packed)
- Regions Covered: Multiple Indian cities including Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Jaipur, Lucknow, Shimla, and many others
- Key Metrics: Total Sales, Average Price, Maximum Price, Minimum Price, Sales by Date, and Product-wise Distribution
- Time Frame: The dataset covers sales transactions over a specific time period, providing trend-based insights
- Format: Structured tabular data imported into Power BI for analysis and visualization

Significance in the Project:

Utilizing this dataset allows for an in-depth analysis of agricultural product performance across different markets. By examining the data, one can identify which products are most popular, which regions generate the highest sales, how prices vary across products, and how sales trends change over time. This information is crucial for farmers, distributors, and policymakers aiming to make informed decisions in the agricultural market.

Data Preprocessing

Before building the dashboard and performing analysis, several data preprocessing steps were carried out in Microsoft Power BI using Power Query Editor to clean and structure the dataset for accurate visualization and interpretation. The dataset was imported in structured tabular format containing agricultural sales information including products, regions, prices, and sales data.

The following preprocessing steps were applied in Power BI:

1. Data Import and Sheet Selection:

The dataset was imported into Power BI and relevant columns were selected based on project requirements. Each column was reviewed to understand its structure and data type, such as product names, region/city names, sales values, prices, and dates.

2. Handling Missing Values:

Missing or blank values were identified using Power Query. Depending on the nature of the data, null values were either replaced with suitable default values (such as zero for sales values) or removed where the data was incomplete or irrelevant.

3. Cleaning and Standardizing Column Names:

Column headers were cleaned by removing unnecessary spaces, symbols, and inconsistencies. Column names were renamed in a consistent format to make the dataset easier to understand and work with.

4. Data Type Transformation:

Numeric columns such as total sales, price values, and date fields were converted into the correct data types (whole number, decimal, or date). This ensured accurate calculations and proper visualization.

5. Data Merging and Relationship Building:

Data from multiple tables or queries was combined where required using merge and append operations in Power Query. Relationships were created between tables using common fields such as product name and region to support interactive analysis.

6. Filtering and Sorting Data:

Unnecessary rows such as totals, subtotals, blank records, and irrelevant entries were removed. The cleaned data was then sorted and organized to improve dashboard readability and performance.

7. Creating Calculated Columns and Measures:

Additional calculated columns and DAX measures were created for KPIs such as total sales, average price, maximum price, minimum price, and percentage share of each product. These measures helped provide deeper insights and support interactive dashboard features.

After preprocessing in Power BI, the dataset was transformed into a clean and structured format, enabling smooth dashboard interactions, accurate analysis, and effective data storytelling.

Objective 1: To Analyze Overall Agricultural Sales Performance

i. General Description

This section of the dashboard provides an overview of the overall agricultural sales performance using key KPI metrics including Total Sales, Average Price, Maximum Price, and Minimum Price. It helps users quickly understand the scale and pricing dynamics of the agricultural market at a glance. This analysis gives a clear picture of the market's performance and supports strategic planning and pricing decisions.

ii. Specific Requirements

- To monitor total sales volume and revenue generated across all products and regions.
- To track average, maximum, and minimum pricing trends for agricultural products.
- To provide a high-level overview of market performance through KPI cards.
- To enable quick comparison of pricing metrics to support procurement decisions.
- To provide stakeholders with an easy-to-understand snapshot of agricultural sales health.

iii. Analysis Results

The dashboard KPI cards reveal the following key metrics for the agricultural sales dataset:

- Total Sales recorded is 13 Million, indicating high volume of agricultural transactions across all products and regions.
- Average Price stands at 34, reflecting the mean pricing across all products and sales transactions.
- Maximum Price recorded is 1.50K, indicating the highest priced transaction in the dataset.
- Minimum Price recorded is 2.00, representing the lowest priced transaction, showing the wide price range across products.

These KPI metrics provide a quick and effective snapshot of the overall market, enabling stakeholders to understand the scale of agricultural commerce and the pricing extremes.

iv. Visualization Used in Dashboard

- KPI Cards: To prominently display Total Sales, Average Price, Maximum Price, and Minimum Price as headline metrics.
- Dynamic Filters / Slicers: To allow users to filter KPIs by product and explore segment-specific performance.

These visualizations in Microsoft Power BI make the overall sales performance easy to interpret and support interactive data exploration.

Objective 2: To Evaluate Product-wise Sales Performance

i. General Description

This section of the dashboard presents a product-wise analysis of total sales, comparing the contribution of each agricultural product such as Groundnut Oil (Packed), Rice, Onion, Brinjal, and Cabbage. The analysis helps in understanding which products dominate the market and which contribute less to the overall sales volume.

This comparison provides useful insights into product demand, market share, and the relative importance of each product in the agricultural supply chain.

ii. Specific Requirements

- To compare the total sales contribution of different agricultural products.
- To identify the best-selling and lowest-selling products.
- To understand the percentage share of each product in overall sales.
- To support procurement, stocking, and supply chain decisions.
- To provide an interactive product filter for focused analysis.

iii. Analysis Results

The dashboard analysis shows clear variation in sales contribution among different products:

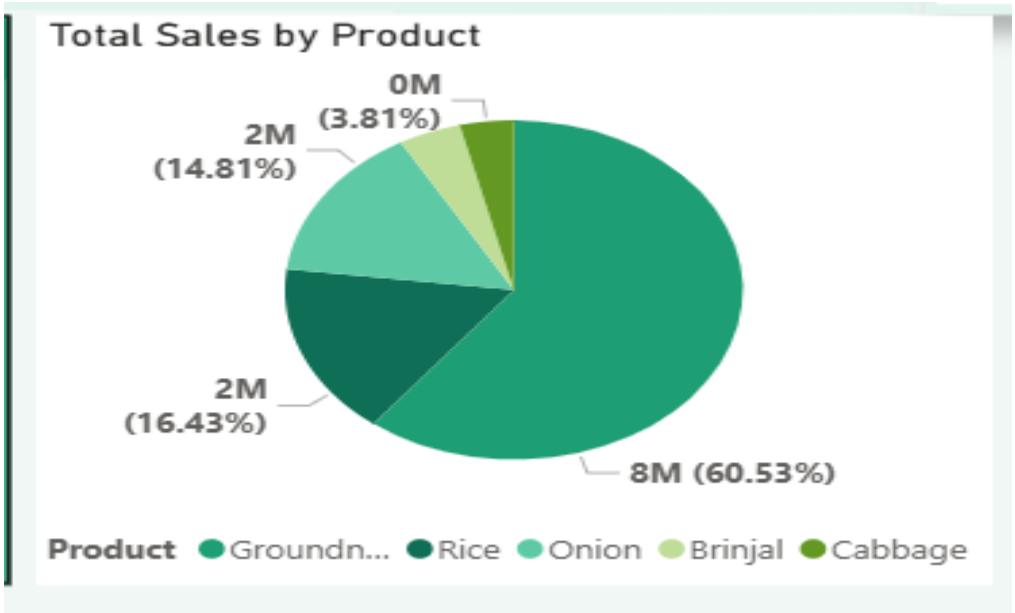
- Groundnut Oil (Packed) dominates with the highest sales share at 60.53% (approximately 8 Million), making it by far the most sold product in the dataset.
- Onion contributes the second-highest share at 16.43% (approximately 2 Million), indicating strong demand for this essential vegetable.
- Rice follows closely with 14.81% (approximately 2 Million), reflecting consistent and widespread consumption.
- Brinjal contributes 3.81% (approximately 0M) and Cabbage makes the smallest contribution, indicating comparatively lower sales volumes.

Overall, the analysis shows that Groundnut Oil (Packed) is the dominant product, while perishable vegetables like Brinjal and Cabbage have comparatively limited sales.

iv. Visualization Used in Dashboard

- Pie Chart: To display the percentage share of total sales contributed by each product.
- Data Labels: To clearly display each product's sales value and percentage contribution.
- Legend: To identify product-wise distribution.
- Interactive Slicer: To allow users to select specific products and filter views.

The pie chart in Microsoft Power BI makes the product-wise sales comparison simple, visually clear, and effective for analysis.



The product slicer panel on the left side of the dashboard allows users to select individual products such as Groundnut Oil (Packed), Rice, Onion, Brinjal, and Cabbage to dynamically filter all charts and visuals. This makes the dashboard highly interactive and supports focused product-level exploration.

Objective 3: To Analyze Region-wise Sales Distribution

i. General Description

This section of the dashboard provides a region-wise analysis of agricultural sales across different Indian cities and regions. It helps identify the top-performing regions and highlights areas with lower sales activity. This analysis is useful for understanding geographic market dynamics, identifying high-demand cities, and supporting regional business strategies.

ii. Specific Requirements

- To identify the highest and lowest sales-generating regions.
- To compare agricultural sales performance across different Indian cities.
- To understand regional disparities in product demand and distribution.
- To support region-level benchmarking and distribution planning.
- To identify areas with scope for improving market reach and sales.

iii. Analysis Results

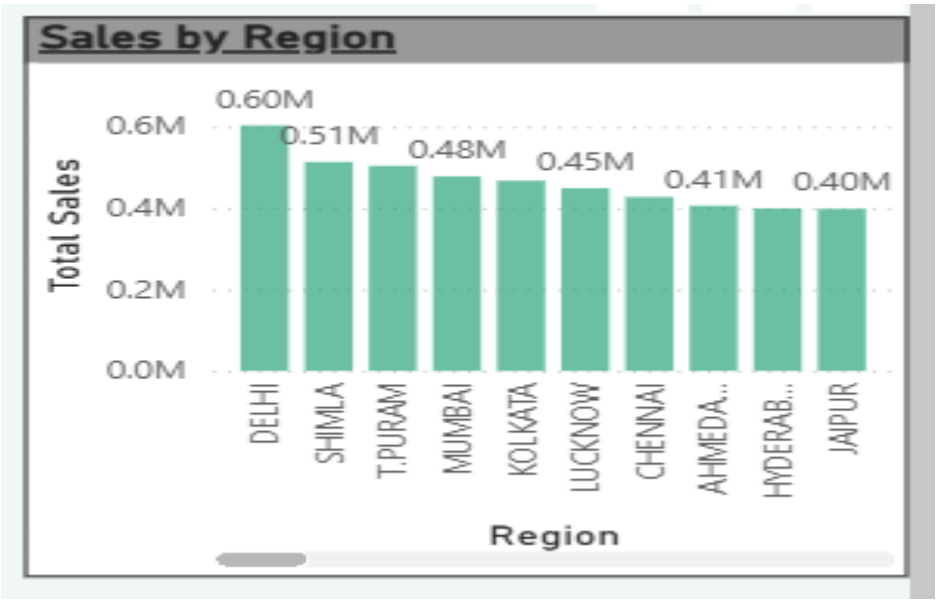
The dashboard analysis shows significant variation in sales performance across regions:

- Delhi recorded the highest total sales at 406,735, making it the top-performing city in agricultural sales.
- Shimla follows closely with 349,238 in total sales, reflecting strong market activity.
- Lucknow, Kolkata, and Mumbai also show strong sales figures ranging from 302,833 to 335,757, indicating high product demand in metro and semi-metro cities.
- Thiruvananthapuram (T.Puram), Ahmedabad, Bhopal, Chennai, and Hyderabad contribute substantial sales, reflecting distributed demand across the country.
- Smaller cities and regions show comparatively lower sales, indicating potential for market expansion.

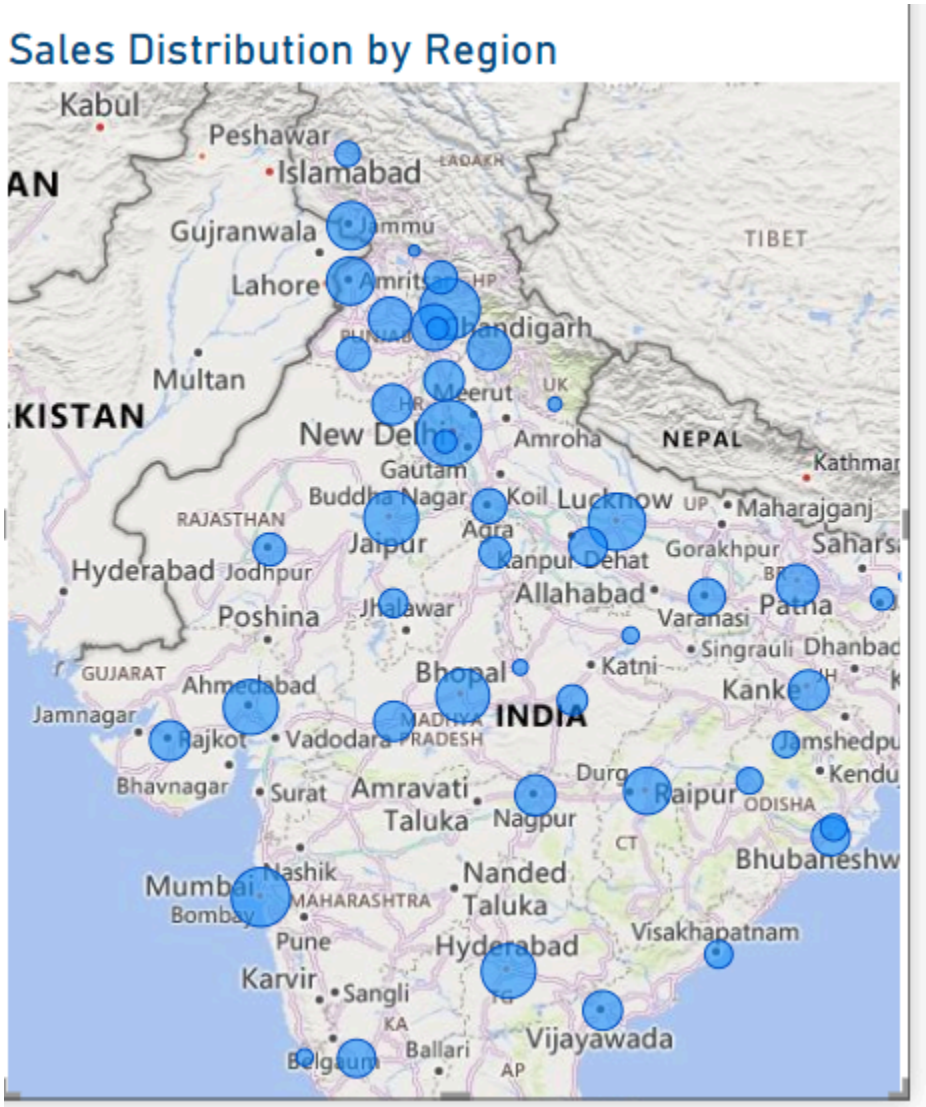
The product-wise sales distribution by region further reveals that Groundnut Oil (Packed) dominates across all major cities, contributing the largest share in every region, while Onion and Rice provide secondary contributions.

iv. Visualization Used in Dashboard

- Bar Chart (Sales by Region): To compare total sales across different cities clearly.
- Stacked Bar Chart (Product-wise Sales Distribution by Region): To show the contribution of each product within each region.
- Map Visual (Sales Distribution by Region): To display geographic distribution of sales across India using bubble markers.
- Data Table: To provide exact sales figures for each region and product combination.



These visualizations in Microsoft Power BI make region-wise agricultural sales analysis easy to understand and support better insights into geographic market performance and product distribution.



Objective 4: To Study Time-based Sales Trends

i. General Description

This section of the dashboard analyzes the total sales generated over different dates to identify patterns such as growth, decline, or seasonal variations in agricultural product sales. It helps in understanding how sales performance changes over time and supports forecasting and inventory planning.

ii. Specific Requirements

- To monitor sales trends over specific time periods.
- To identify patterns of growth, decline, or stability in agricultural sales.
- To detect any seasonal or periodic fluctuations in product demand.
- To support forecasting for future sales and supply planning.
- To provide stakeholders with clear time-series insights for decision-making.

iii. Analysis Results

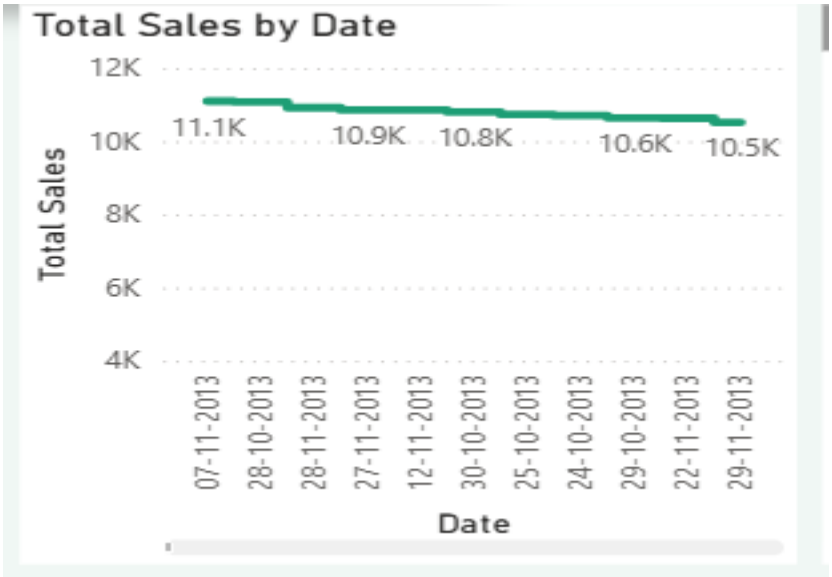
The dashboard time-series analysis reveals the following trends in agricultural sales over the recorded dates (October-November 2013):

- Sales started at the highest point of 11.1K and showed a general declining trend over the recorded period.
- The sales dropped steadily from 11.1K through 10.9K, 10.8K, and 10.6K, reaching 10.5K at the end of the period.
- The decline appears gradual and consistent, suggesting either seasonal demand reduction or market saturation effects during this period.
- The relatively narrow range between maximum (11.1K) and minimum (10.5K) suggests that sales remained moderately stable despite the slight downward trend.

Overall, the analysis shows that agricultural sales were fairly stable over the recorded period, with a slight downward trend. This indicates consistent market activity but with minor fluctuations that could be attributed to supply-demand variations or seasonal factors.

iv. Visualization Used in Dashboard

- Line Chart: To show date-wise trends in total agricultural sales and identify overall patterns.
- Data Labels: To display exact sales figures at key data points along the timeline.
- X-Axis (Date): To clearly represent the time dimension of the analysis.
- Interactive Filters / Slicers: To allow date-specific or product-specific analysis



These visualizations in Microsoft Power BI make time-based agricultural sales trend analysis clear, interactive, and effective for understanding sales patterns and supporting forecasting decisions.

Objective 5: To Enable Interactive & Detailed Data Exploration

i. General Description

This section of the dashboard focuses on providing an interactive and detailed data exploration experience through the use of dynamic filters, maps, charts, and data tables. The Detailed Sales Analysis Dashboard complements the main overview dashboard by enabling users to drill down into specific regions, products, and metrics for deeper analysis.

This interactive approach supports better decision-making by allowing users to explore the data from multiple perspectives and gain customized insights.

ii. Specific Requirements

- To provide dynamic region-based filtering for focused geographic analysis.
- To display Total Records, Max Sales, and Total Regions as KPI metrics on the detailed dashboard.
- To enable product-wise sales distribution analysis across regions through an interactive stacked bar chart.
- To display a geographic map showing sales distribution across Indian cities using bubble markers.
- To present a detailed data table showing region-wise total sales and dominant product for each region.

iii. Analysis Results

The detailed dashboard analysis provides the following key insights:

- Total Records in the dataset: 373K records, indicating a large and comprehensive sales dataset.
- Maximum Sales value recorded: 1.50K, representing the highest single transaction in the dataset.
- Total Regions covered: 372.681K (unique city-product combinations), reflecting the wide geographic coverage of the dataset.
- The region slicer allows users to select specific cities from a list (including Agartala, Agra, Ahmedabad, Aizwal, Amritsar, Bathinda, Bengaluru, and more) to filter all

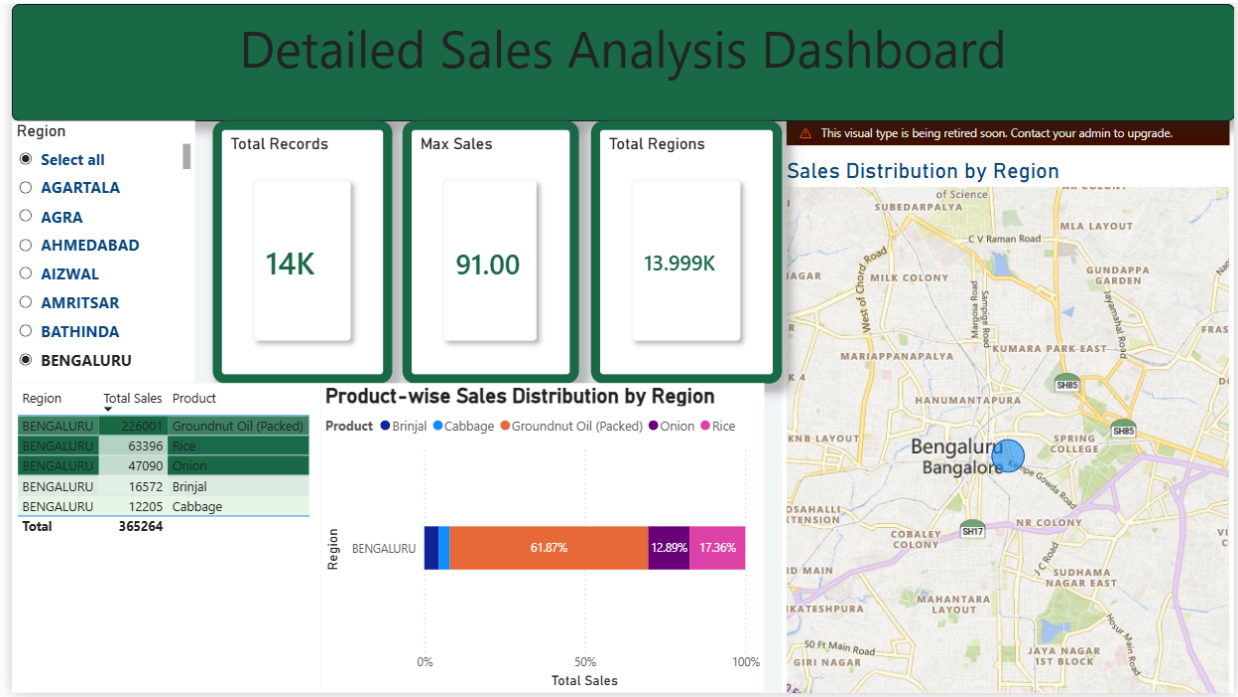
dashboard visuals dynamically.

- The product-wise distribution by region confirms that Groundnut Oil (Packed) contributes the majority share (ranging from 59% to 75%) across all major cities, followed by smaller contributions from Onion, Rice, Brinjal, and Cabbage.
- The geographic map visual provides an intuitive geographic view of sales distribution, with larger bubbles indicating cities with higher total sales.

iv. Visualization Used in Dashboard

- KPI Cards: To display Total Records, Max Sales, and Total Regions as headline metrics on the detailed view.
- Region Slicer: To provide city-level filtering across all dashboard visuals.
- Stacked Bar Chart (Product-wise Sales Distribution by Region): To show proportional product contributions for each city.
- Map Visual (Sales Distribution by Region): To display geographic sales spread using bubble-size encoding across India.
- Data Table: To show exact region-wise Total Sales and top Product for granular data exploration.

These visualizations in Microsoft Power BI make the detailed data exploration interactive, insightful, and easy to use. The dual-dashboard approach (overview + detailed) ensures that users can both get a quick summary and perform deep dives into the data for comprehensive analysis and decision-making.



Conclusion

The analysis of agricultural sales data through the Microsoft Power BI dashboard reveals important insights into the market's performance, product preferences, and regional distribution patterns.

The dashboard shows that the overall agricultural sales reached 13 Million in total, with Groundnut Oil (Packed) emerging as the dominant product contributing over 60% of total sales. This highlights the critical role of packaged edible oils in agricultural commerce.

The region-wise analysis indicates that metropolitan cities like Delhi, Shimla, Lucknow, Kolkata, and Mumbai generate the highest sales volumes, reflecting strong market activity in urban and semi-urban areas. This shows a clear connection between population density, market accessibility, and agricultural product demand.

The time-based trend analysis reveals that sales remained fairly stable over the recorded period with a slight declining trend, suggesting consistent market activity with minor seasonal or demand-driven fluctuations.

The interactive detailed dashboard further demonstrates that Groundnut Oil (Packed) dominates across all regions, while vegetables like Onion and Rice provide moderate contributions. Products like Brinjal and Cabbage have comparatively smaller market shares.

Overall, the project demonstrates that the agricultural sales market is concentrated around a few dominant products and cities, while significant opportunities exist to expand market reach in less-served regions and diversify product portfolios. Future focus should include expanding distribution networks to smaller cities, promoting lesser-sold products, and leveraging data analytics for better demand forecasting and inventory management.

Future Scope

The growing complexity of agricultural markets creates strong opportunities to improve the sales analysis dashboard in the future. Based on the current analysis, the following areas have significant scope for development:

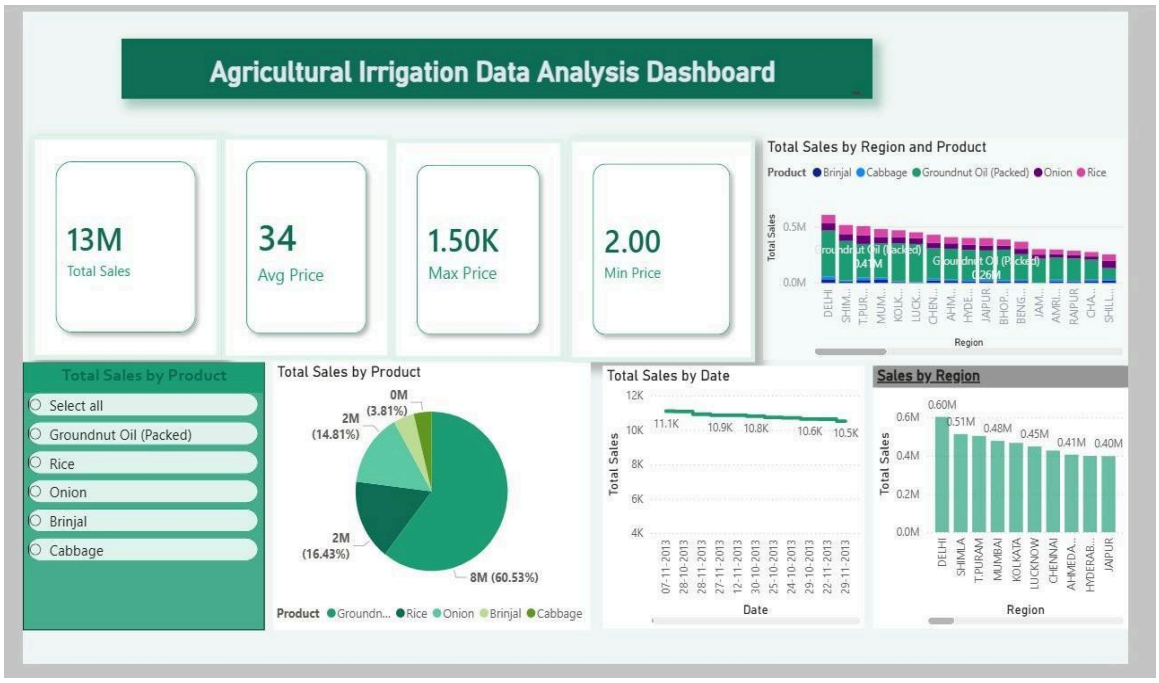
- Expanding Product Coverage: Include more agricultural products such as wheat, pulses, dairy, and spices to provide a more comprehensive market view.
- Longer Time Period Analysis: Extend the dataset to cover multiple years to enable year-on-year comparison and long-term trend analysis.
- Predictive Analytics: Integrate machine learning models for sales forecasting, demand prediction, and price trend estimation.
- Real-time Data Integration: Connect the dashboard to live data sources for real-time monitoring of sales and price fluctuations.
- Regional Expansion: Include rural and semi-urban market data to identify untapped opportunities and support inclusive agricultural development.
- Supply Chain Optimization: Use dashboard insights to improve logistics, reduce waste, and optimize inventory management across the supply chain.
- Sustainability Tracking: Add metrics related to organic products, sustainable farming practices, and eco-friendly packaging to align with future market trends.

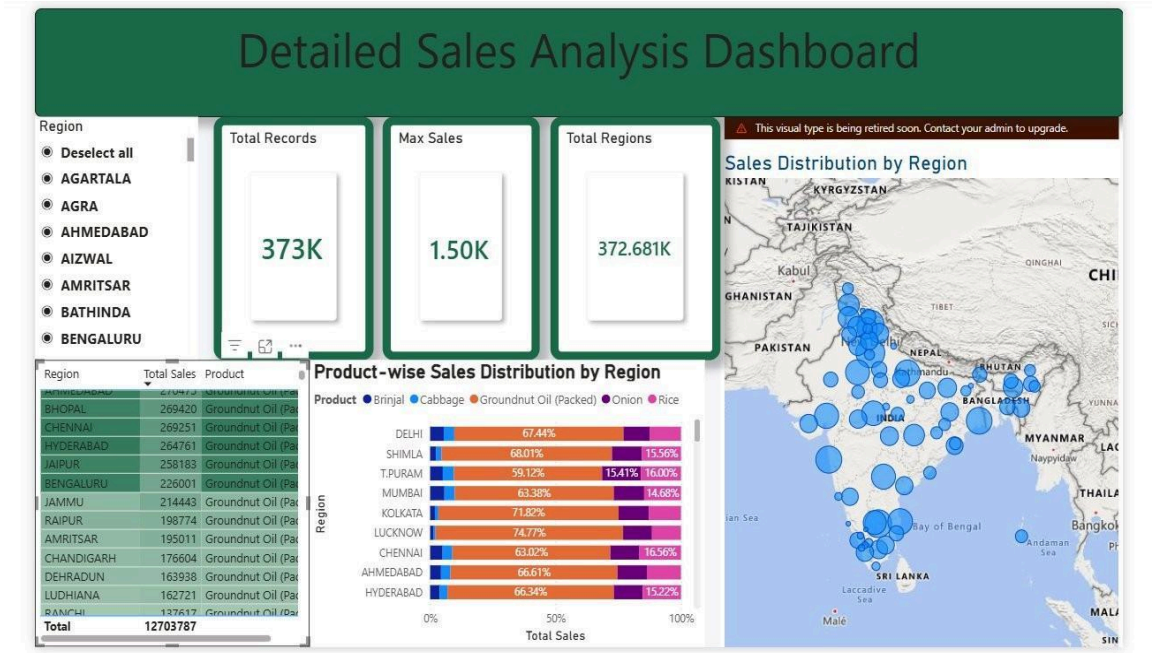
Overall, the agricultural sales dashboard has strong potential to evolve into a comprehensive business intelligence tool that supports farmers, distributors, policymakers, and retailers in making data-driven decisions for a more efficient and sustainable agricultural economy.

References

- □ Open Government Data (OGD) Platform India – Daily Retail Prices of Essential Commodities
- (Multiple Datasets: Rice, Onion, Cabbage, Brinjal, Groundnut Oil (Packed))
- Ministry of Consumer Affairs, Food & Public Distribution, Government of India.
-
- The datasets include region-wise daily retail prices of essential food commodities such as
- Rice, Onion, Cabbage, Brinjal, and Groundnut Oil (Packed) collected across various cities
- in India. These datasets were combined and preprocessed to perform comparative analysis,
- trend analysis, and to build an interactive Power BI dashboard for agricultural price insights.
-
- Retrieved from: https://www.data.gov.in
- https://www.data.gov.in/catalog/dailyweekly-retail-prices-rice
- https://www.data.gov.in/catalog/dailyweekly-retail-prices-cabbage
- https://www.data.gov.in/catalog/dailyweekly-retail-prices-onion

Dashboard


Agricultural Irrigation Data Analysis Dashboard
13M
Total Sales
34
Avg Price
1.50K
Max Price
2.00
Min Price
Total Sales by Region and Product
Product Brinjal Cabbage Groundnut Oil (Packed) Onion Rice
Total Sales
0.5M
0.0M
Region
Total Sales by Product
Select all
Groundnut Oil (Packed)
Rice
Onion
Brinjal
Cabbage
Total Sales by Product
0M (3.81%)
2M (14.81%)
2M (16.43%)
8M (60.53%)
Product Groundn... Rice Onion Brinjal Cabbage
Total Sales by Date
12K
10K
8K
6K
4K
11.1K
10.9K
10.8K
10.6K
10.5K
Total Sales
Date
Sales by Region
0.60M
0.51M
0.48M
0.45M
0.41M
0.40M
0.6M
0.4M
0.2M
0.0M
Total Sales
DELHI
SHIMLA
T.PURAM
MUMBAI
KOLKATA
LUCKNOW
CHENNAI
AHMEDA...
HYDERAB...
JAIPUR
Region


Detailed Sales Analysis Dashboard
Region
Deselect all
AGARTALA
AGRA
AHMEDABAD
AIZWAL
AMRITSAR
BATHINDA
BENGALURU
Total Records
373K
Max Sales
1.50K
Total Regions
372.681K
This visual type is being retired soon. Contact your admin to upgrade.
Sales Distribution by Region
Region Total Sales Product
BHOPAL 269420 Groundnut Oil (Pa
CHENNAI 269251 Groundnut Oil (Pa
HYDERABAD 264761 Groundnut Oil (Pa
JAIPUR 258183 Groundnut Oil (Pa
BENGALURU 226001 Groundnut Oil (Pa
JAMMU 214443 Groundnut Oil (Pa
RAIPUR 198774 Groundnut Oil (Pa
AMRITSAR 195011 Groundnut Oil (Pa
CHANDIGARH 176604 Groundnut Oil (Pa
DEHRADUN 163938 Groundnut Oil (Pa
LUDHIANA 162721 Groundnut Oil (Pa
Total 12703787
Product-wise Sales Distribution by Region
Product Brinjal Cabbage Groundnut Oil (Packed) Onion Rice
DELHI 67.44%
SHIMLA 68.01% 15.56%
T.PURAM 59.12% 15.41% 16.00%
MUMBAI 63.38% 14.68%
KOLKATA 71.82%
LUCKNOW 74.77%
CHENNAI 63.02% 16.56%
AHMEDABAD 66.61%
HYDERABAD 66.34% 15.22%
Region
0%
50%
100%
Total Sales
KYRGYZSTAN
TAJIKISTAN
Kabul
PAKISTAN
NEPAL
BHUTAN
BANGLADESH
INDIA
MYANMAR
Bay of Bengal
SRI LANKA
Laccadive Sea
Malé

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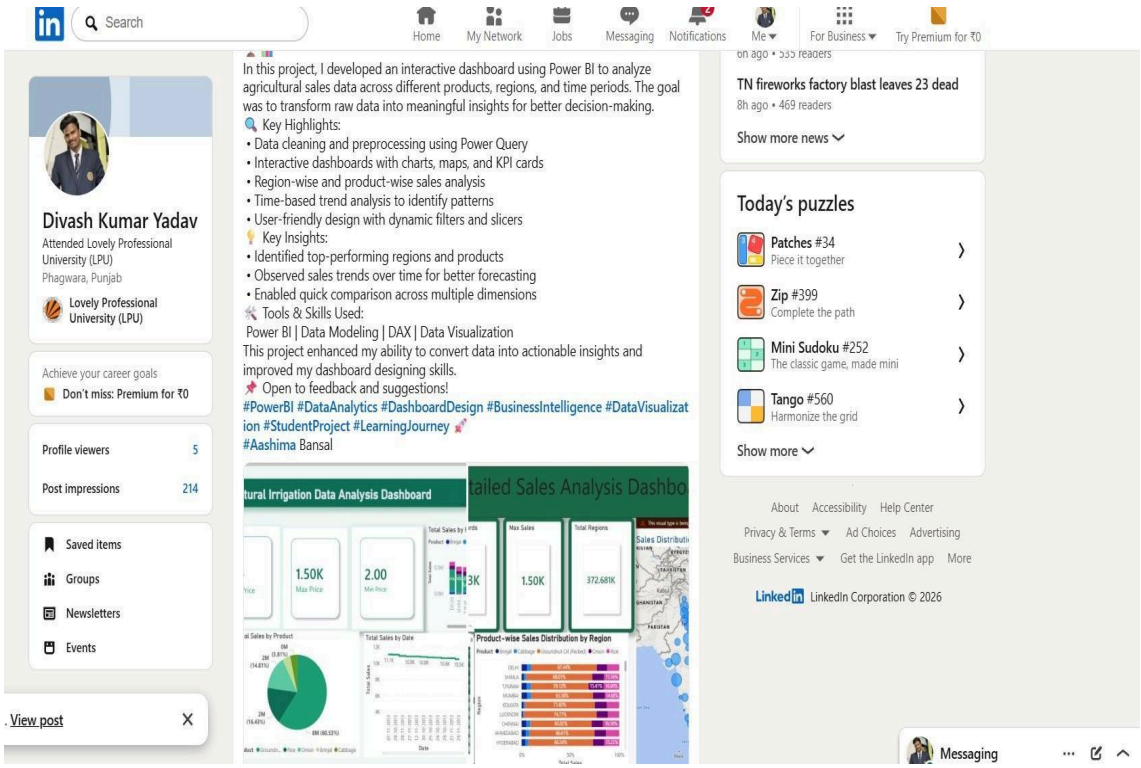

GITHUB REPOSITORY:

URL:https://github.com/divash45/Power-BI-Project